Mapped seasonal surface water decline under changing observation support in northwestern Bangladesh

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Abstract

Changes in satellite-mapped water can inform climate-resilient management only when observation support is explicit. We assessed annual seasonal and permanent water source classes in Naogaon, Chapainawabganj and Rajshahi, northwestern Bangladesh, using JRC histories and gridded climate data. The primary 2000–2021 analysis used a retrospectively selected, annually observed footprint of 1,276.41 km². Exploratory reanalysis compared seven footprint/validity cohorts, multi-year windows, simpler climate models, approximate uncertainty estimates and chronological hindcasts. Seasonal source-class area fell from 938.53 to 699.91 km² between the endpoints (25.42%); comparison of the first and final five-year means gave a smaller decline of 14.70%. A footprint based on water observed in 2000–2004 and annual validity through 2021 gave an 18.79% five-year decline. Negative seasonal coefficients persisted across the tested models, although interval and district support varied. Permanent-class gains were sensitive to observation and processing controls. Static river and 2020 cropland context separated contrasting mapped trajectories without identifying their causes. During 2015–2024, raising the Dynamic World threshold from 0.5 to 0.7 increased conditional radar–optical agreement from 89.04% to 98.71%, while mean paired support fell from 11.07% to 3.29%; wet-season support was particularly sparse. A two-year joint-cell pilot showed that high-threshold comparisons contained almost exclusively water-class area. Assisted interpretation of 720 sampled pixel-months assigned 76.54% of design-weighted area-time, with 98.56% conditional agreement but no independent accuracy estimate. No actual drought-recovery duration was identified under nine support/reference specifications. Synthetic pulse diagnostics retaining real gaps and envelope widths also failed to confirm the tested deficits. The results identify a restricted-footprint seasonal decline and the observation requirements for investigating it; district-wide water storage, causal abstraction effects and independently validated resilience remain unmeasured.

Keywords: seasonal surface water; climate adjustment; observation support; drought monitoring; remote sensing uncertainty; Barind Tract; Bangladesh

1 Introduction

Seasonal surface water connects rainfall variability with the timing and location of usable water. For water managers, a contraction in mapped seasonal water can indicate a place requiring investigation, but its interpretation depends on what the satellite actually observed. A water map summarizes a classification process, a spatial footprint and a set of acquisition dates. Changes in any of these can affect the apparent hydrological signal. The Landsat-based Global Surface Water record provides a consistent framework for examining long-term water occurrence and annual source classes (Pekel et al., 2016). Its application to local water-management questions nevertheless requires attention to missing observations, class definitions and updates to the underlying archive.

These concerns are relevant to northwestern Bangladesh. The Barind region has pronounced seasonal water stress and substantial dependence on groundwater-supported agriculture. Recent station-based research documented spatially heterogeneous rainfall trends across Naogaon, Chapainawabganj and Rajshahi during 2000–2024 (Esha et al., 2026). Work in the high Barind has also examined heatwaves, rainfall, vegetation, soil moisture and groundwater jointly (Sarker et al., 2026). These studies motivate a surface-water assessment that retains local differences and distinguishes measured water classifications from possible climatic and management mechanisms. Administrative districts provide reporting units for such an assessment, although their boundaries do not define hydrological catchments.

Existing evidence also cautions against assuming one direction of surface-water change throughout Bangladesh. In the southwestern Bengal delta, a Landsat time-series study examined strong seasonality and reported expansion in mapped surface water alongside changing local water management (Huda et al., 2021). Those findings concern another region and observation design. They cannot establish the trajectory of seasonal water in the northwest, but they show why the period, water class and spatial domain of a trend need to be explicit. At the much larger Indus–Ganges–Brahmaputra scale, hydrological modeling has separated the components contributing to total water-storage change (Prusevich et al., 2025). Surface-water class area alone does not provide that decomposition.

Combining sensors offers additional evidence, but agreement between products is conditional on where both can assign labels. Dynamic World provides Sentinel-2-derived land-cover probabilities at 10 m (Brown et al., 2022). Radar can supply observations under cloud conditions that restrict optical mapping, while remaining sensitive to vegetation, surface roughness and the water-detection method. A multi-site comparison of Sentinel-1 and Sentinel-2 explicitly separated open and vegetated water and evaluated both against higher-resolution imagery (Vanderhoof et al., 2023). In Senegal, field, satellite and modeled pond observations revealed both the value and the limitations of remotely sensed water availability (Slinski et al., 2025). These examples support reporting complementary evidence without treating another satellite product as error-free truth. Research on index-based flood insurance in Bangladesh further shows that data choice can affect the timing and uncertainty of water-related decisions (Saunders et al., 2025).

Drought resilience introduces a stricter temporal requirement. A rainfall deficit can propagate through water stores with delays and persistence, so the end of meteorological drought need not coincide with hydrological recovery (Van Loon, 2015). Estimating a recovery duration requires observations before, during and after the event. A complete collection of monthly files can still lack the continuous, interpretable pixel observations needed for this calculation. Similarly, a large number of human image interpretations does not establish independent accuracy when the reviewed samples were selected by a model or the reviewer saw model suggestions. Sampling design, reference quality and analysis must be considered together (Olofsson et al., 2014; Stehman and Foody, 2019).

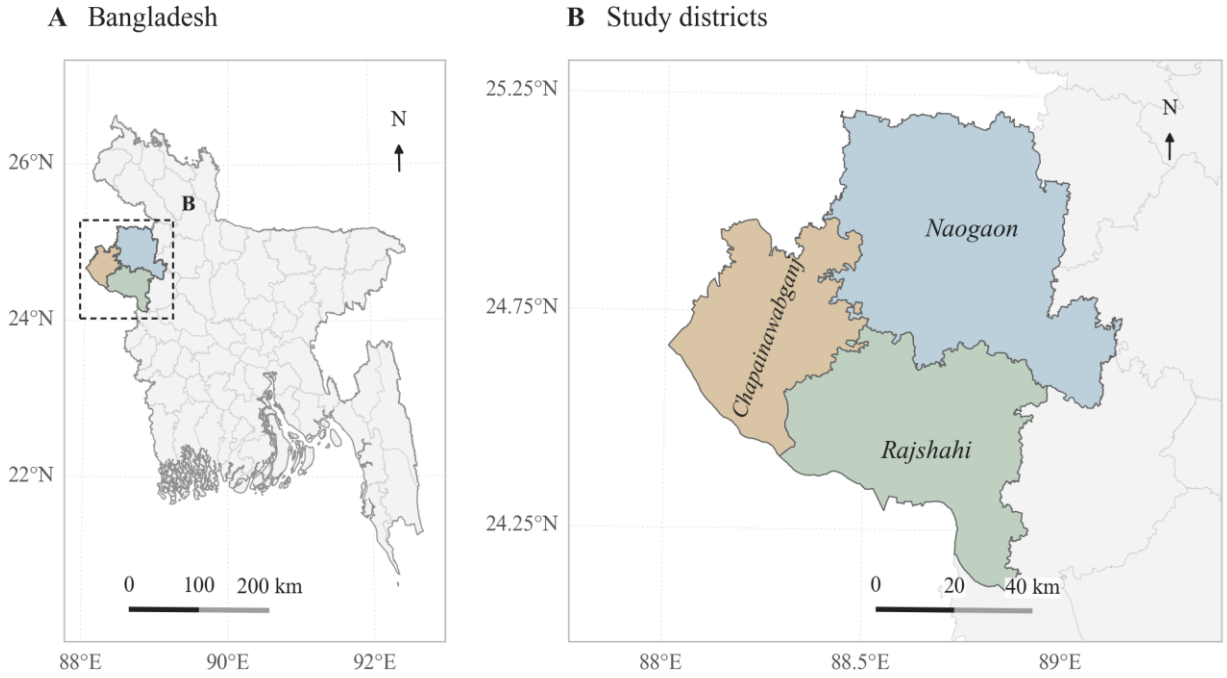
We therefore ask three related questions: how did seasonal and permanent source-class areas change within a fixed observed footprint; which annual associations persist after climate, observation-effort and processing-boundary controls; and how much spatial and temporal evidence remains available for sensor comparison and drought-recovery assessment? The contribution is an integrated regional assessment with explicit limits on its estimands, rather than a new classification algorithm. We retain null results, unsupported months and unresolved interpretations throughout. This provides a practical basis for

deciding which mapped changes merit local investigation and which resilience claims require additional observations.

2 Materials and methods

2.1 Study area and analytical design

The study covers Naogaon, Chapainawabganj and Rajshahi in northwestern Bangladesh (Figure 1). The reporting boundary was obtained from geoBoundaries Bangladesh ADM2, boundary identifier BGD-ADM2-16705992, representing 2020 and citing Bangladesh Bureau of Statistics and OCHA ROAP as source organizations. The archived source names Chapainawabganj as Nawabganj; we harmonized its reporting name while preserving the geometry and source identifiers. Boundary metadata, licensing and download identifiers are retained in the reproducibility materials. District estimates and their union are reported separately; the union overlaps all three districts and is not an independent replicate.



C Analysis domains

Component	Period	Evidence and analytical scope
Annual source classes	2000–2021	JRC seasonal and permanent classes Fixed footprint: 1,276.41 km²; climate and observation controls
Radar–optical comparison	2015–2024	Sentinel-1 and Dynamic World; separate retrospective cohorts Threshold and recovery-gap sensitivity
Dated image comparison	Four selected months	720 stratified pixel-months; dated Sentinel-2 imagery Assisted interpretation; proxy agreement; unknowns retained

Separate analysis domains; satellite agreement is not independent accuracy.

Figure 1. Study location and analytical design. (A) Location of the study districts within Bangladesh. (B) The three district reporting boundaries; Chapainawabganj is harmonized from the source name Nawabganj. (C) The primary annual analysis, separate radar–optical analysis and four-month dated comparison. Maps use a WGS84 transverse Mercator projection centered on 90°E; scale bars and north arrows are geodesic. Boundaries: geoBoundaries gbOpen BGD ADM2, BGD-ADM2-16705992; underlying sources Bangladesh Bureau of Statistics and OCHA ROAP, CC BY 3.0 IGO. Boundaries are shown for analysis and imply no position on legal status.

The design has three linked but distinct components. The primary component describes annual Joint Research Centre (JRC) source classes during 2000–2021 within a fixed annual-observation footprint. A 2000–2024 extension assesses source-version sensitivity. A complementary component uses Sentinel-1 and Dynamic World during 2015–2024 on separately constructed historical water cohorts. Finally, a stratified sample in four selected months compares JRC binary labels with dated Sentinel-2 proxies and assisted human interpretation. These components differ in period, spatial support and outcome; their trend estimates are not pooled.

2.2 Data sources, versions and preprocessing

Table 1 summarizes the data. Water and climate products were exported from Google Earth Engine (Gorelick et al., 2017), then audited locally for source periods, duplicate records, masks, units, grid metadata and checksums. JRC annual classes were retained as no observations (0), nonwater (1), seasonal water (2) and permanent water (3). Monthly classes were retained as no observations (0), nonwater (1) and water (2). Masked export values were nodata, not evidence of dry land. Pixel areas and climate-polygon intersections in the primary local analysis were evaluated in the equal-area coordinate system EPSG:6933, while native source grids were preserved.

Table 1. Data sources and their roles. Resolution denotes product/grid spacing, rather than independently resolved hydrological detail. Components use different spatial domains.

Source	Period used	Product spacing	Purpose and limitation
JRC Global Surface Water	2000–2021 primary; 2022–2024 sensitivity	30 m	Annual source classes and monthly observations; corrected 2016–2021 assets and a collection change after 2021
CHIRPS version 3	1981–2024	0.05°	Rainfall and fixed-baseline SPI; complete six-pentad monthly sums
ERA5-Land monthly aggregated	2000–2024	Approximately 0.1° grid	Temperature and model-specific evaporation demand; no measured groundwater or reference ET_0
Sentinel-1 GRD	2015–2024	10 m nominal product grid	One descending orbit; monthly VV benchmark classes and method envelopes
Dynamic World version 1	2015–2024; availability from June 2015	10 m	Four confidence thresholds; unclassified support retained
Sentinel-2 Level-2A	Four sampled months	10 m display; B11/SCL native 20 m	Actual dated crop values and assisted interpretation; not independent ground truth
geoBoundaries gbOpen ADM2	Represented year 2020	Vector district polygons	Fixed reporting boundaries; administrative districts are not catchments
HydroRIVERS Asia version 1	Static network	15 arc-second source hydrography	River-proximity context; not measured channel width or discharge
ESA WorldCover 2020 version 100	2020	10 m	Static cropland context; modal class on JRC grid; not historical land-use change

Annual data use the historical JRC collection through 2015, corrected 2016–2021 YearlyHistory assets, and separate 2022–2024 assets. The provider reports that the recent extension uses Landsat Collection 2, whereas the earlier record uses Collection 1. Residual co-registration differences can be spatially variable and can reach or exceed one 30 m pixel. The provider also documents corrections to 2016–2021

seasonality and yearly-history layers.[1] Consequently, the primary trend period ends in 2021. The extended analysis includes a 2022 source-version indicator, and a separate pre-2022 sensitivity includes a 2016 corrected-asset indicator. Neither indicator establishes that a coincident physical change is an artifact.

Rainfall was obtained from CHIRPS version 3 pentads at 0.05° (Funk et al., 2026). A monthly value required all six actual pentads to be valid at a pixel; annual rainfall required all twelve monthly values. Regional climate summaries used exact cell-intersection areas and finite-data support. ERA5-Land monthly aggregated fields supplied 2 m temperature and potential evaporation (Muñoz-Sabater et al., 2021). Kelvin was converted to degrees Celsius, accumulated fluxes from meters to millimeters, and annual temperature was weighted by calendar-month days. The original signed evaporation flux and its sign-reversed demand proxy were stored separately. ERA5-Land potential evaporation is a model-specific evaporative-demand measure; it was not interpreted as FAO reference evapotranspiration.[2]

2.3 Fixed footprints, observation effort and annual outcomes

The retrospective reference footprint contains pixels classified as seasonal or permanent water in at least one year during 2000–2024. The annual comparison footprint is its subset with a valid annual class in every one of those 25 years. Although primary outcomes stop in 2021, footprint construction therefore uses subsequent observation availability and water classification. This restriction is intentional and retrospective; it does not represent an independently defined wetland inventory or an operational forecasting domain. Within each region and year, seasonal and permanent areas are sums of pixel areas assigned the corresponding annual source class on that fixed footprint.

A separate transition diagnostic compared strict five-year majority classes in 2000–2004 and 2017–2021 on the same annual footprint. All five annual observations were required in each window, and at least three occurrences defined the majority class. A window without a strict majority remained ambiguous. These transitions describe source-class assignments and do not validate land-use or storage change.

Observation-effort summaries include the area-weighted number of observed calendar months, the number observed during June–September, and the area fraction observed in all twelve months. These diagnostics distinguish annual class availability from a fully observed annual water cycle. Permanent source classification is not treated as independent confirmation that water was visible in every month. For the original monthly JRC analysis, candidate months required coverage thresholds of 80%, 90% or 95% on both the district raster domain and the retrospective footprint. The common monthly cohort intersects valid pixels across all candidate months. A retained outcome required complete observation on that cohort, at least 48 candidate months and at least 20% footprint retention. All support failures remain in Supplementary Data.

Initial seasonal and permanent cohorts required all five annual observations in 2000–2004 and at least three occurrences of the respective class. Ambiguous and initially nonwater pixels were excluded from these two cohorts. Original monthly JRC inference for initial-class cohorts began in 2005, while the all-water series began in 2000. A monthly detection on an initial permanent cohort does not relabel that cohort as a new permanent class each month. The all-water common footprint retained 2.46%, 17.36% and 50.35% of its reference area under the 80%, 90% and 95% candidate-month gates, respectively. These rules select different restricted cohorts; their differences are not a coverage-treatment effect.

2.4 Annual trend, climate and observation models

Unadjusted annual class areas were regressed on year; descriptive Theil–Sen slopes were also retained. Climate models used the logit of each class's share of the fixed footprint. With year measured from 2000, the annual model is given in Equation (1):

$$\operatorname{logit}(p_t) = \alpha + \beta_t(t - 2000) + \beta_P P_t + \beta_T T_t + \beta_E E_t + \boldsymbol{\gamma}^{\top}\boldsymbol{O}_t + \delta I(t \geq 2016) + \varepsilon_t. \qquad (1)$$

[1] JRC Global Surface Water data access and update notes, accessed 16 October 2026: https://global-surface-water.appspot.com/download.

[2] ECMWF ERA5-Land documentation: https://confluence.ecmwf.int/spaces/CKB/pages/140385202/ERA5-Land+data+documentation. Earth Engine monthly aggregation: https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_MONTHLY_AGGR.

Here, P_t is annual rainfall, T_t is day-weighted mean temperature, E_t is annual sign-reversed potential evaporation, and $\boldsymbol{O}_t$ contains observation-effort terms. The baseline climate model omitted observation and processing terms. Three exploratory sensitivities added, sequentially, mean annual observed months; annual plus June–September observed months; and those observation terms plus the 2016 indicator. The final specification contains both observation terms and the processing-boundary indicator. The 2000–2024 sensitivity uses the relevant 2022 version indicator separately.

Fractions were clipped to $[10^{-6}, 1-10^{-6}]$ solely for logit fitting; stored observations were unchanged. Predictors were centered and scaled during fitting, with coefficients transformed back to original units. Rank and condition-number checks rejected unusable designs. Conditional 95% coefficient intervals used Student-t critical values and finite-sample-corrected Bartlett heteroskedasticity- and autocorrelation-consistent covariance, with annual lag 2 (Newey and West, 1987). Lag matching used actual calendar distances. District-level p-values were adjusted by Benjamini–Yekutieli within each annual class-trend or climate-model family, and across the six district-by-class tests within each observation-sensitivity specification (Benjamini and Yekutieli, 2001). The union was descriptive and excluded from those district testing families. Intervals shown in figures are pointwise; these families do not correct for every exploratory choice in the study.

Model acceptance required enough usable observations beyond the number of parameters, finite and nonconstant features, full design rank and a scaled condition number below 10^8. The rank and condition checks assess numerical usability rather than correctness of the regression specification.

The time coefficient is a conditional change in logit class share per year. It is not an area-loss rate or an estimate of human water use. Fixed-climate displays hold weather at regional 2000–2021 medians. Separate modeled seasonal and permanent shares are not summed into a compositional prediction. Temporal validation used two-year calendar test blocks beginning in 2009, 2013, 2017 and 2020, requiring at least twelve earlier training years and two test years for annual fits. Thus, three annual folds were accepted. Feature estimation was confined to training data. A training-only mean class fraction provided the annual benchmark. The resulting errors assess conditional hindcast performance on the retained footprint, not water-classification accuracy.

Original monthly checks used one annual sine/cosine harmonic, three-calendar-month rainfall, monthly temperature and evaporation demand. Monthly folds required 36 training observations and six test observations, with the preceding three calendar months excluded from training. Scaling was estimated within each training set, and the monthly benchmark used training calendar-month means. Every accepted and rejected fold and null model outcome remains in the extended diagnostic tables.

2.5 Radar–optical analysis and support-sensitive agreement

The 2015–2024 radar–optical analysis reconstructed its own retrospective annual ever-water footprint on a fixed approximately 30 m angular grid. Two additional frozen cohorts required all five initial annual classes to be observed and a strict majority of seasonal or permanent source classes during 2000–2004. These historical cohorts retained their initial labels throughout the radar analysis. They are distinct from the annual comparison footprint.

Sentinel-1 GRD scenes were restricted to interferometric wide swath, VV/VH availability, incidence angles of 30–45° and one pass/relative-orbit track. A deterministic metadata-based ranking selected descending relative orbit 48 before water-result comparisons. Selection considered a footprint proxy from at most the first six scenes, then temporal scene availability, scene count and lexical ties; it did not measure valid pixel coverage. Actual support was evaluated subsequently. Input scenes were placed on a common EPSG:32645 10 m working grid. Scenes on the same acquisition date were mosaicked, and a monthly pixel required at least two unique acquisition dates. Median VV, already in decibels, was classified at −19, −17 and −15 dB. These are strict, central and loose benchmark specifications without local independent calibration; no additional speckle filter was applied.

For track ranking only, the metadata footprint calculation simplified the boundary by 1,000 m and quantized the footprint proxy to 0.001; actual water summaries retained the original boundary. Dynamic

World land evidence used trees, grass, crops, shrub_and_scrub, built, bare and snow_and_ice. Component-wise temporal medians of water and maximum-land scores were not assumed to form one probability distribution summing to one. The mean paired-footprint fraction includes all 120 months, including zero-support months, whereas pooled agreement divides summed agreeing area-time by summed paired area-time.

For Dynamic World, daily mosaics supplied monthly median water probability and median maximum probability among the seven selected land classes. Flooded vegetation was excluded from the land maximum. At least two unique dates were required. At each threshold $q \in \{0.5,0.6,0.7,0.8\}$, water evidence required water probability at least q; nonwater evidence required land probability at least q and water probability at most $1 - q$. Other pixels remained unclassified. These model scores were not treated as calibrated local probabilities of correctness. Dynamic World shares Sentinel-2 optical lineage with the dated interpretation component, so the optical comparisons are dependent (Brown et al., 2022).

Overlap-weighted supported and water fractions were aggregated from the working grid to the fixed coarse grid. A coarse comparison cell required at least 95% supported footprint and a water fraction of at least 80% or at most 20% of its supported area. Agreement was the summed agreeing area-time divided by summed paired area-time; retained paired footprint was reported alongside it. Monthly composites may represent different acquisition dates between sensors, adding genuine within-month change to classification and registration differences. No confidence threshold was selected by maximizing agreement.

A two-year joint-cell pilot retained all twelve months of 2018 and 2024, four regions (three districts and their union), three historical-water cohorts and all four confidence thresholds. The eight authenticated quarter exports comprised 288 region/cohort/month rows and used the frozen scene snapshot and radar track. Each product pair was partitioned into both-water, first-water/second-nonwater, first-nonwater/second-water and both-nonwater areas. The first/second products were SAR/Dynamic World and Dynamic World/JRC, respectively. Symmetric positive agreement was twice both-water area divided by twice both-water plus discordant area; negative agreement used twice both-nonwater in the corresponding ratio. Product-specific water prevalence used the same paired denominator. Monthly areas were summed as area-time (km²-month); support denominators retained every intended month, including zero-support months. Empty denominators remained unavailable. These descriptive product comparisons do not estimate sensitivity, specificity or independent physical-water accuracy.

Radar method envelopes retained both threshold and observation uncertainty. If D is fixed cohort area, V is supported area and W_{19} and W_{15} are water areas under the strict and loose VV specifications, respectively, Equation (2) defines the envelope:

$$[L, U] = \left[\frac{W_{19}}{D}, \frac{W_{15} + D - V}{D}\right]. \qquad (2)$$

The upper endpoint allocates unsupported area to water. This range does not encompass all physical classification errors and is not a confidence interval for true water area. Exploratory endpoint regressions included time, one annual sine/cosine harmonic, three-month rainfall, temperature, evaporation demand and, where variable, radar coverage and acquisition-day count. Calendar-distance HAC lag 12 and separate district testing families for lower and upper endpoints were used. All region/cohort results, including unsupported findings, were retained.

2.6 Drought definition and recovery identifiability

SPI-3 and SPI-6 were calculated from complete-calendar rolling CHIRPS rainfall totals, using a fixed 1981–2020 calibration. Calendar-month-specific gamma maximum likelihood fixed location at zero and estimated zero probability separately, followed by a normal-quantile transformation. Calibration required at least 30 usable accumulations and ten positive accumulations for each calendar month. Failed fits

remained missing. This is a precipitation-only drought index, without a validated precipitation–evapotranspiration index.[3]

An SPI-3 event began at ≤−1 and ended at the first subsequent month with SPI ≥0; missing climate data censored active events. For radar recovery, eligible monthly envelopes required at least 95% supported area. Calendar-month reference envelopes were estimated from eligible 2015–2021 observations, with at least five reference observations. Subtracting the upper reference from the current lower endpoint, and the lower reference from the current upper endpoint, gave a conservative anomaly envelope. The pre-event reference required three consecutive observed months. A detectable drought deficit had to exceed 0.02 of cohort area; recovery required a return within 0.02 of the pre-event anomaly for two consecutive observed months, with up to 24 months of follow-up.

For numerical stability, cumulative probabilities were bounded away from 0 and 1 at 10^{-8}. In the recovery calculation, the drought deficit excludes the first non-drought termination month. All event records retain their original missing-support status. Neither repeated dates within a sample nor overlapping event–cohort records are independent replicates.

Possible and confirmed recovery times were sought separately from the anomaly endpoints. Missing pre-event, drought-phase or follow-up evidence prevented a duration estimate. No interpolation bridged gaps. Sensitivity crossed coverage gates of 80%, 90% and 95% with minimum calendar-month reference counts of three, four and five. The resulting event–cohort records overlap across districts, union and cohorts. They are accounting records, not independent drought replicates. Original JRC recovery diagnostics were retained separately because their calendar reference and observation domains differ.

2.7 Dated sample comparison and assisted interpretation

The frozen comparison frame comprised March 2018, November 2020, March 2023 and November 2024. Within each district, month and observed JRC binary class, 30 pixels were sampled without replacement, giving 24 strata and 720 pixel-months. Previously exposed review cells were excluded before this new sample was drawn. The design therefore applies to this eligible four-month population, rather than every pixel or year in the annual study. The frozen sampling seed and stratum populations are retained in the source records.

Actual Sentinel-2 Level-2A crop values and product XML calibration supported an initial spectral triage. The rules used modified normalized difference water index (Xu, 2006), normalized green–near-infrared difference, vegetation index and reflectance screens. At least 95% of the target footprint had to be clear and strongly classified on at least two dates spanning fourteen days. A boundary diagnostic flagged opposite spectral evidence in a 10 m surrounding ring. These deliberately restrictive heuristic rules were not trained or locally validated as an accuracy model. Although displays used a 10 m grid, the shortwave-infrared band and scene-classification layer retained their coarser native information content.

The spectral indices used in the triage are defined in Equation (3), where G, N, R and S denote green, near-infrared, red and shortwave-infrared reflectance, respectively:

$$\begin{aligned} \mathrm{MNDWI} &= (G - S)/(G + S), \\ \mathrm{NDWI} &= (G - N)/(G + N), \\ \mathrm{NDVI} &= (N - R)/(N + R). \end{aligned} \qquad (3)$$

Only finite reflectance within [0,1.2] was screenable, and invalid or near-zero denominators remained unavailable. Accepted scene-classification values were 4, 5 and 6; this operational clear-pixel screen does not exclude every residual cloud or shadow error. Strong water required MNDWI ≥0.20, NDWI ≥0.10, NDVI ≤0.20, near-infrared reflectance ≤0.15, shortwave-infrared reflectance ≤0.08, green reflectance ≥0.01 and mean visible reflectance ≥0.012. Strong nonwater required either MNDWI ≤−0.20 and NDWI ≤−0.10 with near-infrared or shortwave-infrared reflectance ≥0.08; or NDVI ≥0.45, near-infrared reflectance ≥0.18, MNDWI ≤0, NDWI ≤−0.10 and shortwave-infrared reflectance ≥0.08.

[3] SPI guidance: https://climatedataguide.ucar.edu/climate-data/standardized-precipitation-index-spi.

At least 95% of target area had to pass both the clear-pixel and strong-class requirements for a dated rule label. More than 5% radiometrically uninterpretable target area prevented acceptance. Fractions used the actual target footprint intersected with crop cells. The surrounding 10 m ring triggered a boundary flag if at least half of the ring was clear and at least 10% of its area had the opposite weak index sign. Monthly automatic assignment required at least two accepted dates spanning fourteen days, no conflicting or ambiguous dated rule and no boundary flag. Raw crops comprised five reflectance layers, scene classification and support on a 40×40 display grid; each reflectance band used its actual product XML scale and offset.

Difficult cases selected by the triage were reviewed date by date with image assistance. A monthly comparison label was retained only when its accepted dated interpretations were consistently binary; temporal changes, uncertain dates and inadequate support remained unresolved. Repeated dates were evidence within one pixel-month, not additional independent samples. One dated answer was subsequently revised from nonwater to water at the reviewer's direction after the rule output was disclosed. Original and revised files were preserved. Accordingly, this component measures assisted proxy agreement, without claiming blinded independent validation. The thresholds and revision procedure are specified above; Supplementary Methods retain the exact revision identifiers.

Human dated answers were joined by sample_id and scene_id, never by display row. The original export contained 429 water, 237 nonwater and 28 uncertain answers. The documented correction changed one answer from nonwater to water, yielding the current totals reported below. Its timestamp records a revision after rule output was shown, not a new independent image inspection. The original CSV, backup and cell-difference record remain preserved; exact product and sample identifiers are retained in Supplementary Methods, Section S2.

For sampled pixel i in stratum h, the design weight was $w_i = (N_h/30)a_i$, where N_h is the eligible population pixel count and a_i is sampled geodesic pixel area. Normalized weights approximate area-time ratios; they are not a full census. Let A denote assigned samples and M assigned samples matching JRC. We report assigned weight, conditional agreement $\sum_{i\in M} w_i \,/\, \sum_{i\in A} w_i$, and the binary-completion range in Equation (4):

$$\left[\frac{\sum_{i\in M} w_i}{\sum_i w_i}, \frac{\sum_{i\in M} w_i + \sum_{i\notin A} w_i}{\sum_i w_i}\right]. \qquad (4)$$

This range permits arbitrary agreement or disagreement for unresolved labels. It bounds completion of the proxy comparison, not physical-water accuracy. Approximate sampling intervals used 2,000 pixel-month percentile bootstrap replicates within the original strata, with seed 20261016. No finite-population correction or spatial-cluster model was used. These intervals exclude reference, timing, radiometric, registration and spatial-clustering errors and do not resolve nonrandom reference failure. Degenerate conditional bootstrap intervals remained unavailable rather than being presented as certainty.

2.8 Exploratory robustness and landscape context

A revision protocol was frozen before the new sensitivity outputs were examined; the original study results were already known. Seven overlapping cohorts were retained: C1, the original 2000–2024 ever-water footprint with all 25 annual classes valid; C2, the same ever-water condition with validity required only through 2021; C3, ever-water and complete annual validity during 2000–2021; C4, any seasonal/permanent class in 2000–2004, all five baseline classes valid, and complete annual validity through 2021; C5, the same baseline-water condition with available observations each year; C6, 2000–2021 ever-water with at least twenty valid annual classes; and C7, all annually valid pixels through 2021 within the district boundaries, without an ever-water restriction. C7 is still an observed subset, not full-district coverage. C1–C7 union areas were 1,276.41, 1,276.41, 1,271.76, 1,196.72, 1,252.36, 1,472.73 and 1,299.60 km², respectively. Available-support analyses required at least 95% annual coverage; missing-area bounds were retained rather than classifying gaps as nonwater.

For each cohort, first-to-final one-, three- and five-year windows complemented whole-series trends. A multi-year comparison required every year in both windows to pass its support rule. A separate monthly diagnostic intersected valid observations for each calendar month across all 22 primary years within the baseline-water footprint; a trend required retention of at least 20% of that footprint. This diagnostic measures the cost of equal calendar-month support and cannot reconstruct missing seasons.

The annual model sequence included time alone; time plus June–September rainfall; those terms plus annual observed months, wet-season observed months and the 2016 indicator; an additional preceding October–May rainfall term; and the original annual-climate control specification in Equation (1). Each class was modeled separately. Fits required at least five residual degrees of freedom. Calendar-distance HAC lags zero and two, all leave-one-year-out fits, and a short 2013–2021 time/monsoon analysis were retained. Approximate fixed-design circular residual block bootstrap intervals used lengths two, three and four, 2,000 replicates and seed 20261019. Bootstrap intervals were unavailable for nonconsecutive selected years; gaps were not compressed. District BY families comprised six district-by-class tests per cohort/model. These overlapping sensitivities are not independent confirmations.

New chronological checks used two-year tests starting in 2013, 2017 and 2020, with at least twelve earlier training observations. Mean and trend-only benchmarks were fitted on the same accepted training/test folds as each candidate. Pooled RMSE was the square root of total test squared error divided by test count. A constant processing indicator invalidated earlier training designs, leaving only two folds/four test years for controlled models, compared with three folds/six years for the monsoon model. These retrospective conditional hindcasts neither validate water classification nor establish operational forecasting performance.

HydroRIVERS Asia version 1 centerlines supplied static river context (Lehner and Grill, 2013), and ESA WorldCover 2020 version 100 supplied a static 10 m cropland class (Zanaga et al., 2021). River buffers were formed in EPSG:32645 at 200, 500 and 1,000 m; 500 m was the designated central setting. WorldCover classes were transferred by modal aggregation to the native JRC grid. Within each cohort, river proximity and, away from rivers, 2020 cropland versus other cover formed exhaustive groups. Buffer ranges are sensitivity ranges, not confidence intervals. This context does not identify individual ponds, historical cultivation, irrigation abstraction or river discharge.

A missingness-aware synthetic diagnostic placed rectangular deficits of zero, 0.01, 0.02, 0.05, 0.10 and 0.20 of cohort area at every monthly onset during 2015–2024, for durations of one, three and six months. A zero anomaly baseline returned abruptly at the pulse end, so the imposed recovery time was zero months. The diagnostic retained actual observation gaps, anomaly-envelope half-widths and calendar-reference eligibility, requiring three pre-event observations and two recovery confirmations. It tested detectability under those widths, not a physical drought model or empirical recovery distribution. No actual missing water observations or reference labels were filled.

2.9 Reproducibility and analytical scope

The original computational audit verified 9,393 source-file checksums and reconciled export inventories. The revision additionally rechecked all fifty annual/monthly source rasters, reconstructed the original annual summaries to within 10^{-6} km², and reran the explicitly frozen exploratory models. The corrected dated-label version and its history were unchanged. Protocols, source versions, output tables, failed fits and SHA-256 manifests accompany the code. An initial calendar-gap bootstrap implementation was corrected before the reported outputs were frozen; superseded intervals are excluded. The environment used Python 3.12, NumPy 2.3.5, pandas 2.3.3, statsmodels 0.14.5 and rasterio 1.5.0. Computational checks do not establish environmental accuracy or confirmatory inference.

3 Results

3.1 Seasonal source-class decline on a restricted footprint

The fixed annual footprint was 1,276.41 km², retaining 79.30% of the retrospective ever-water reference area. District contributions were 473.82 km² in Naogaon, 358.88 km² in Chapainawabganj and 443.71

km² in Rajshahi. These areas define the estimand for the primary annual results; they do not cover the complete districts.

Between 2000 and 2021, union seasonal source-class area decreased from 938.53 to 699.91 km², a loss of 238.62 km² or 25.42% relative to the initial seasonal class area (Figure 2; Table 2). The largest district endpoint loss was in Naogaon: 145.96 km², or 37.84%. Rajshahi lost 73.38 km², or 24.46%, and Chapainawabganj lost 19.28 km², or 7.62%. The unadjusted union seasonal linear slope was −8.42 km² year^{-1} (95% interval −12.79 to −4.05; 22 annual observations). Endpoint differences and the fitted slope summarize different aspects of the trajectory.

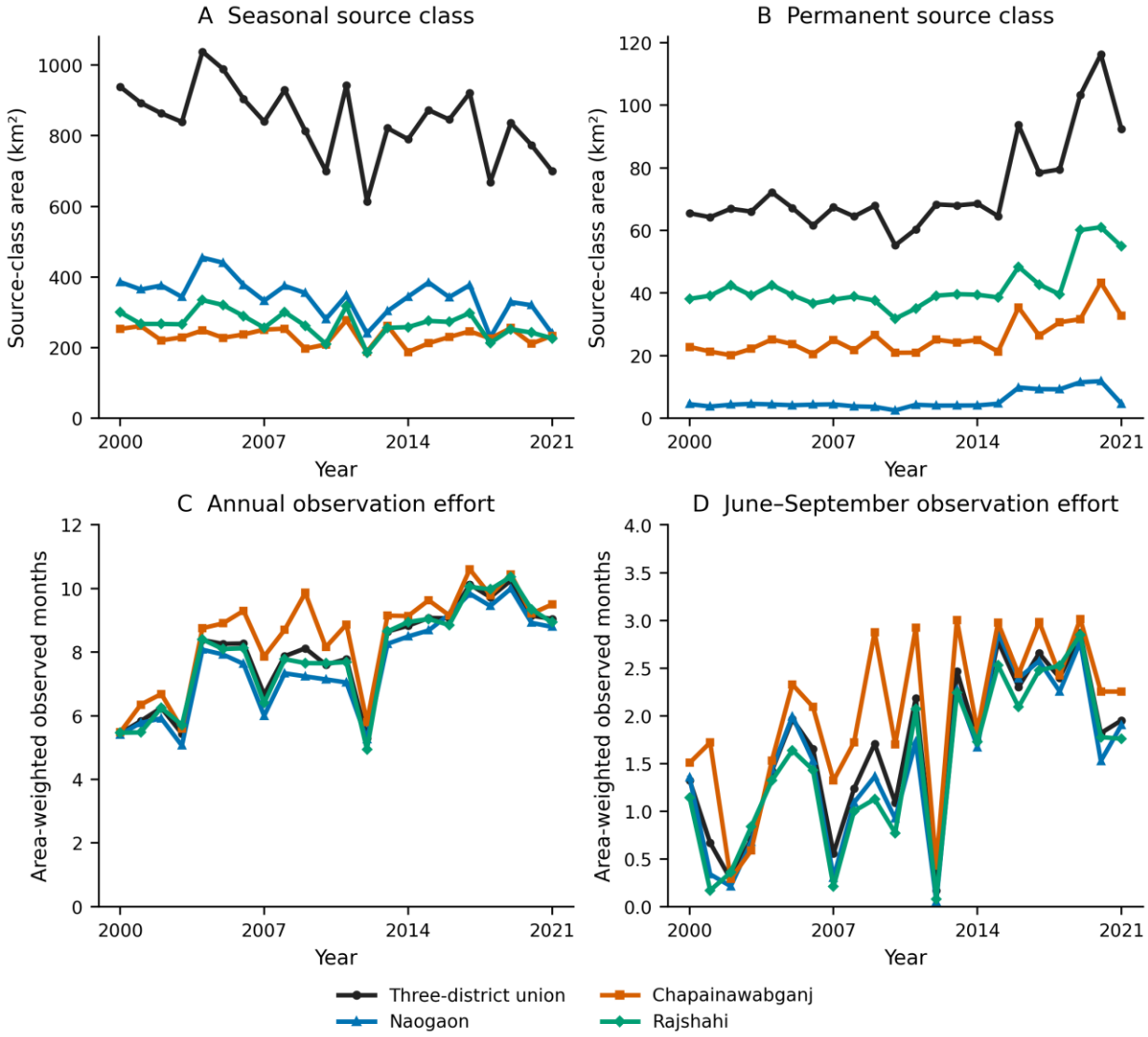


Figure 2. Annual source classes and observation effort on the fixed 1,276.41 km² annual footprint during 2000–2021. (A) Seasonal-class area. (B) Permanent-class area, with its own vertical scale. (C) Area-weighted observed months out of twelve. (D) Observed June–September months out of four. The union overlaps the district series. Annual classes do not certify complete observation of the year. Source: EC JRC/Google; derived from the frozen annual class and observation-quality tables.

Table 2. Annual source-class endpoint changes within the fixed annual-observation footprint, 2000–2021. Percent change is relative to each class's 2000 area. Districts sum to the union; the union is not an independent replicate. These are classified areas, not water volumes or independently validated hydroperiod areas.

Region	Source class	2000 (km²)	2021 (km²)	Change (km²)	Change (%)
Union	Seasonal	938.53	699.91	-238.62	-25.42
Union	Permanent	65.49	92.37	+26.88	+41.04
Naogaon	Seasonal	385.72	239.76	-145.96	-37.84
Naogaon	Permanent	4.52	4.66	+0.14	+3.08
Chapainawabganj	Seasonal	252.84	233.56	-19.28	-7.62
Chapainawabganj	Permanent	22.81	32.80	+9.99	+43.79
Rajshahi	Seasonal	299.98	226.60	-73.38	-24.46
Rajshahi	Permanent	38.16	54.91	+16.75	+43.90

Permanent source-class area increased from 65.49 to 92.37 km² across the same endpoints, a gain of 26.88 km² or 41.04%. Most of that endpoint increase occurred in Rajshahi and Chapainawabganj; Naogaon's permanent-class area changed by only 0.14 km² (Table 2). These descriptive gains do not establish continuous permanent-water expansion after accounting for observation and processing changes.

Figure 3 separates gross majority-class transitions from static landscape context. Among pixels with a majority in both windows, 264.1 km² moved from seasonal to nonwater and 110.1 km² from nonwater to seasonal; seasonal-to-permanent and permanent-to-seasonal transitions were 29.2 and 32.8 km². A further 11.04 km² lacked a strict majority in at least one window. At the 500 m river buffer, the change in five-year mean seasonal area was −64.49 km² near mapped rivers, −89.74 km² in more distant 2020 cropland and +19.84 km² in the remaining distant area. Their initial seasonal-area changes were −15.46%, −26.91% and +12.12%. Across the 200–1,000 m buffers, the respective percentage ranges were −15.54 to −15.25, −27.82 to −26.12 and +7.22 to +20.62. The partition changes with buffer distance; neither these ranges nor source-class transitions establish causes.

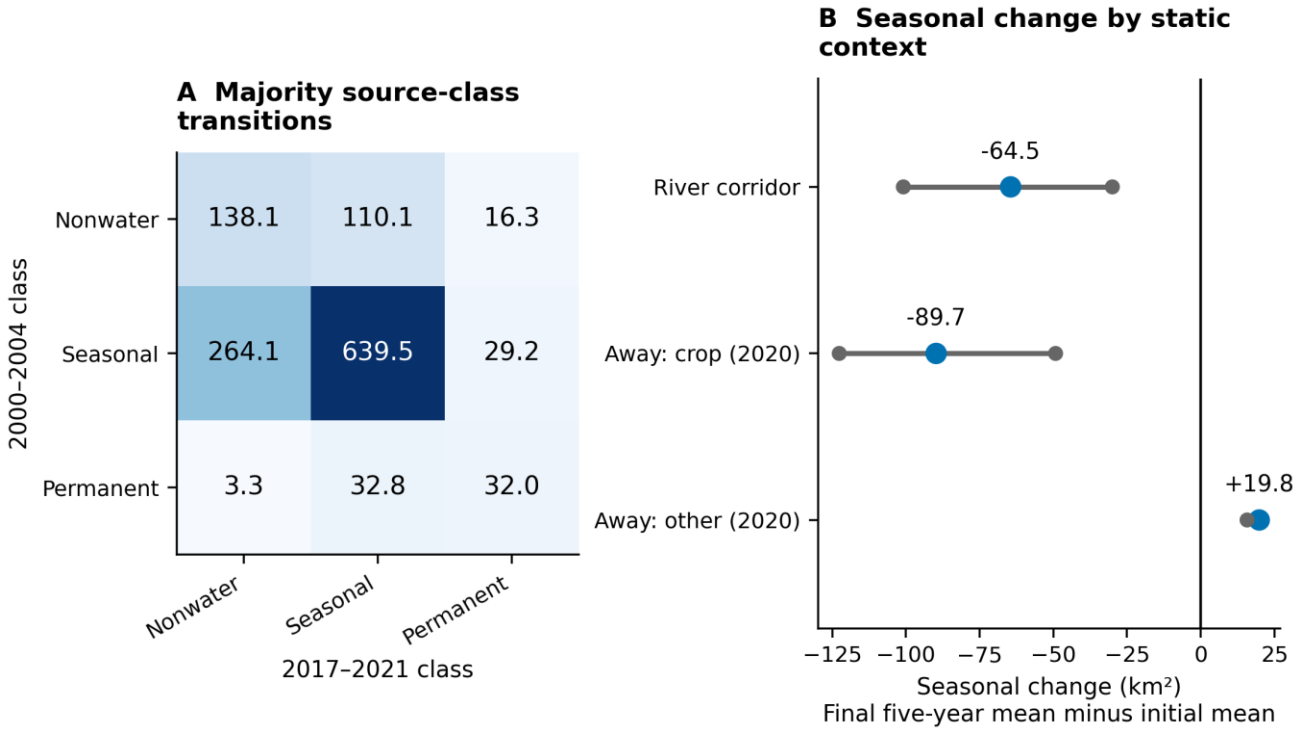


Figure 3. Gross source-class transitions and static landscape context on the original annual footprint. (A) Area in km² between strict five-year majority classes in 2000–2004 and 2017–2021; 11.04 km² without a majority in either window are excluded from the matrix. (B) Change between first and final five-year

mean seasonal areas. Blue dots use a 500 m HydroRIVERS buffer; grey extents cover 200–1,000 m buffer settings and are sensitivity ranges, not uncertainty intervals. Away from the river buffer, groups use static WorldCover 2020 cropland or other cover. Context and class transitions do not establish historical land conversion or causal drivers.

3.2 Observation and processing controls change the permanent-water interpretation

Annual class availability concealed substantial variation in within-year observation effort. On the union footprint during 2000–2021, the area-weighted observed-month count ranged from 5.30 to 10.25 months per year. June–September effort ranged from 0.17 to 2.87 of four months. The footprint fraction observed in every calendar month ranged from 0% to 16.15% (Figure 2). Thus, the annual dataset did not provide a fully observed twelve-month water cycle over most of the retained area.

Seasonal time coefficients remained negative in the union across the observation-control specifications (Figure 4). With climate, annual and wet-season observation effort, and the 2016 indicator included, the union coefficient was −0.0607 logit share year^{-1} (95% interval −0.0814 to −0.0399). District coefficients were −0.1154 in Naogaon (−0.1602 to −0.0706; adjusted p=0.00109), −0.0480 in Rajshahi (−0.0714 to −0.0245; adjusted p=0.00454) and −0.0350 in Chapainawabganj (−0.0720 to 0.0020; adjusted p=0.302). The district distinction is material: the final specification supported a continuing seasonal decline in two districts, while uncertainty included zero in the third.

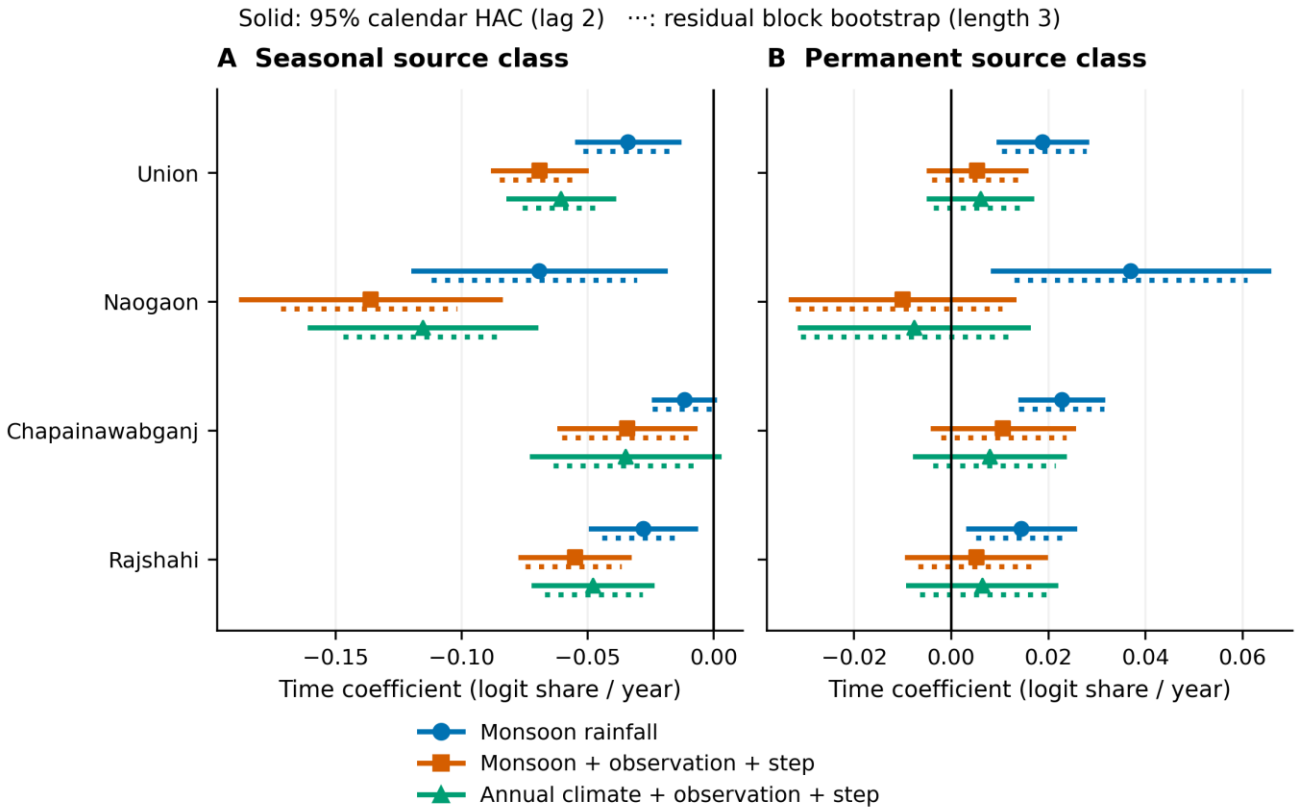


Figure 4. Annual time-coefficient sensitivity to simpler climate and observation controls on the original footprint, 2000–2021. (A) Seasonal class. (B) Permanent class. Models use June–September rainfall alone with time; rainfall plus annual/wet-season observation counts and a 2016 indicator; or the original annual rainfall, temperature, evaporation, observation and processing specification. Solid bars are pointwise 95% calendar-distance HAC lag-two intervals; dotted bars are approximate 95% fixed-design circular residual block bootstrap intervals of length three. Each fit uses 22 annual observations. The union overlaps districts; district multiplicity is reported separately in the text and tables.

All permanent-class time-coefficient intervals included zero after the 2016 processing-boundary indicator was added. The union coefficient was 0.0060 logit share year^{-1} (−0.0046 to 0.0165). Positive permanent trends from simpler specifications were therefore sensitive to this control. A step indicator can absorb a

genuine hydrological discontinuity as well as a processing-related one, so this result does not identify the source of that sensitivity.

Simpler models retained the direction of the seasonal association. In the union, the monsoon-rainfall model gave −0.0341 logit share year^{-1} (95% HAC interval −0.0541 to −0.0140); adding both observation terms and the 2016 indicator gave −0.0691 (−0.0875 to −0.0507). Its residual block intervals were −0.0855 to −0.0528, −0.0850 to −0.0541 and −0.0835 to −0.0555 for lengths two, three and four. District BY-adjusted p-values for this reduced controlled model were 0.000407 in Naogaon, 0.0734 in Chapainawabganj and 0.000407 in Rajshahi. A pointwise interval excluding zero in Chapainawabganj did not imply significance in that adjusted family. The 2013–2021 trend-only union seasonal interval included zero (−0.0431; −0.0887 to 0.0025); adding monsoon rainfall gave −0.0896 (−0.1608 to −0.0183). Short-series and overlapping-cohort results are sensitivities, not independent replication.

The new models and matched-fold aggregation changed the hindcast comparison. Union seasonal pooled RMSE was 0.0745 for the monsoon model versus 0.0899 for the mean and 0.0628 for the trend benchmark on six test years. The reduced controlled model gave 0.0486 versus matched mean/trend errors of 0.1039/0.0726 on four years. Adding antecedent rainfall gave 0.0458, while the original full-control specification gave 0.0691 on those four years. These small, differing test sets limit generalization. The original climate-only analysis and its mean-of-fold-RMSE summaries remain archived and are distinct from this matched-fold pooled comparison.

Figure 5 compares the footprint, time-window, equal-calendar support and hindcast sensitivities. On C1, the first-to-final three- and five-year seasonal changes were −14.28% and −14.70%, compared with −25.42% for endpoints. The five-year decrease was −18.79% on C4 and −19.72% on C5; C6 lacked sufficient coverage for a five-year comparison. All accepted primary seasonal point coefficients were negative across the tested cohorts/models, but their intervals and dependence preclude treating them as independent confirmations. Only November retained at least 20% of the union baseline-water footprint under the 22-year same-calendar rule (25.35%); June–October retained none. Equalizing calendar-month support therefore did not yield a representative complete annual cycle.

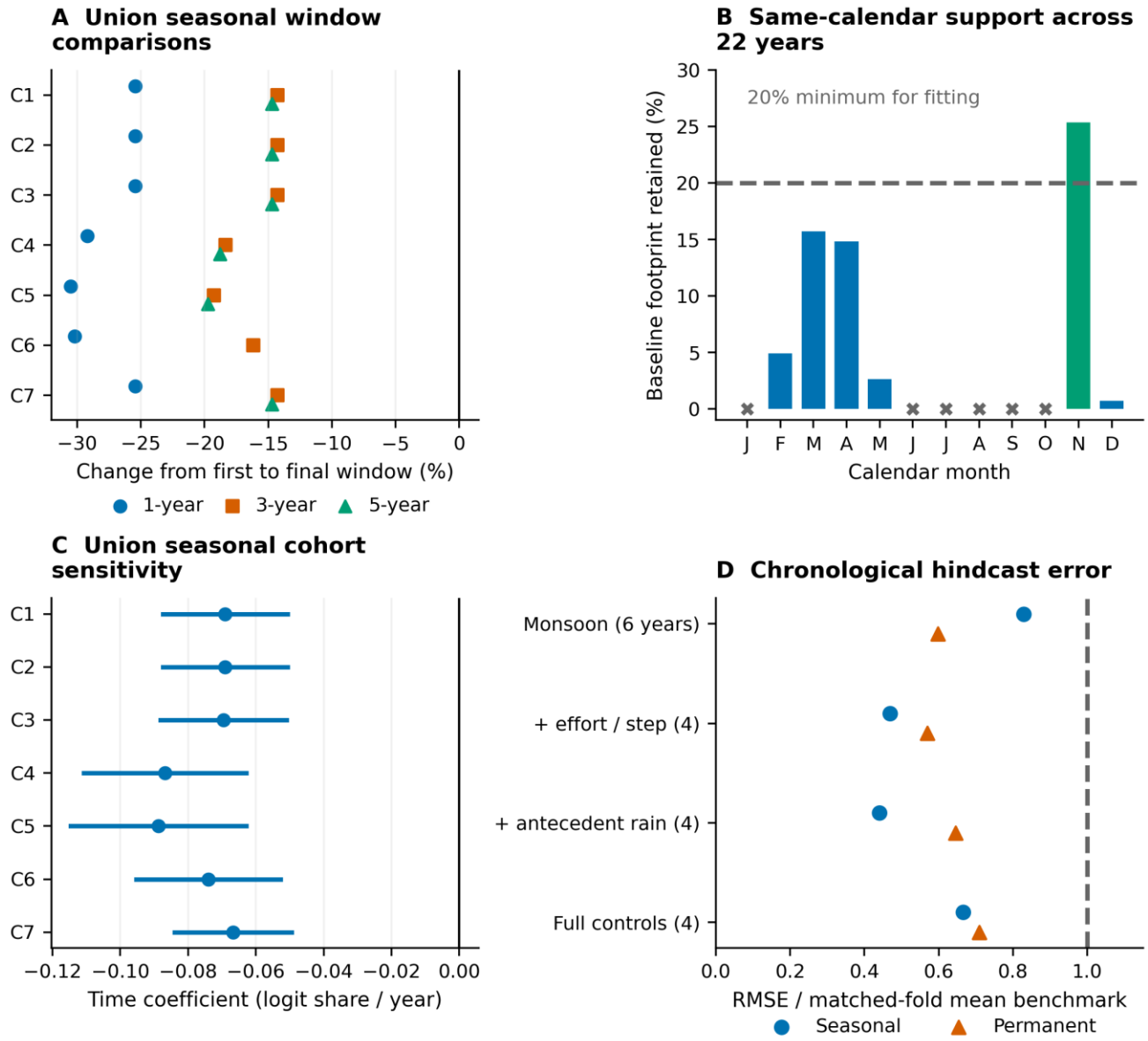


Figure 5. Sensitivity to comparison windows, footprint definition and temporal validation. (A) Union seasonal changes for one-, three- and five-year windows, 2000–2021. C1–C7 are defined in Section 2.8; a missing symbol indicates an unavailable comparison. (B) Baseline-water footprint retained by identical calendar-month observations across all 22 years; crosses show zero support and the dashed line is the 20% fitting gate. (C) Seasonal coefficients and pointwise 95% HAC lag-two intervals for the monsoon/observation/processing model across cohorts. (D) Pooled chronological test RMSE divided by the training-only mean benchmark on exactly the same accepted folds; test-year counts are in parentheses. A ratio below one indicates lower error than that benchmark, not water-classification accuracy.

3.3 Higher automated agreement coincides with lower paired support

The radar–optical analysis reconciled 1,440 monthly region/cohort rows and 5,760 threshold-specific rows. Its scene inventory contained 265 Sentinel-1 scenes and 1,274 Dynamic World scenes, totaling 1,539 unique source identifiers. These complete exports did not imply complete pixel observations in every month.

For the union historical ever-water cohort, increasing the Dynamic World threshold from 0.5 to 0.7 reduced the number of months with any classified support from 76 to 74 of 120. Mean classified support fell from 17.94% to 4.45% of the fixed cohort. Conditional Sentinel-1–Dynamic World agreement increased from 89.04% to 98.71%, while mean paired support decreased from 11.07% to 3.29% (Figure

6). At 0.6, agreement was 95.94% on mean paired support of 5.88%. Threshold 0.8 yielded no classified or paired support. Its missing agreement value indicates absence of comparison evidence.

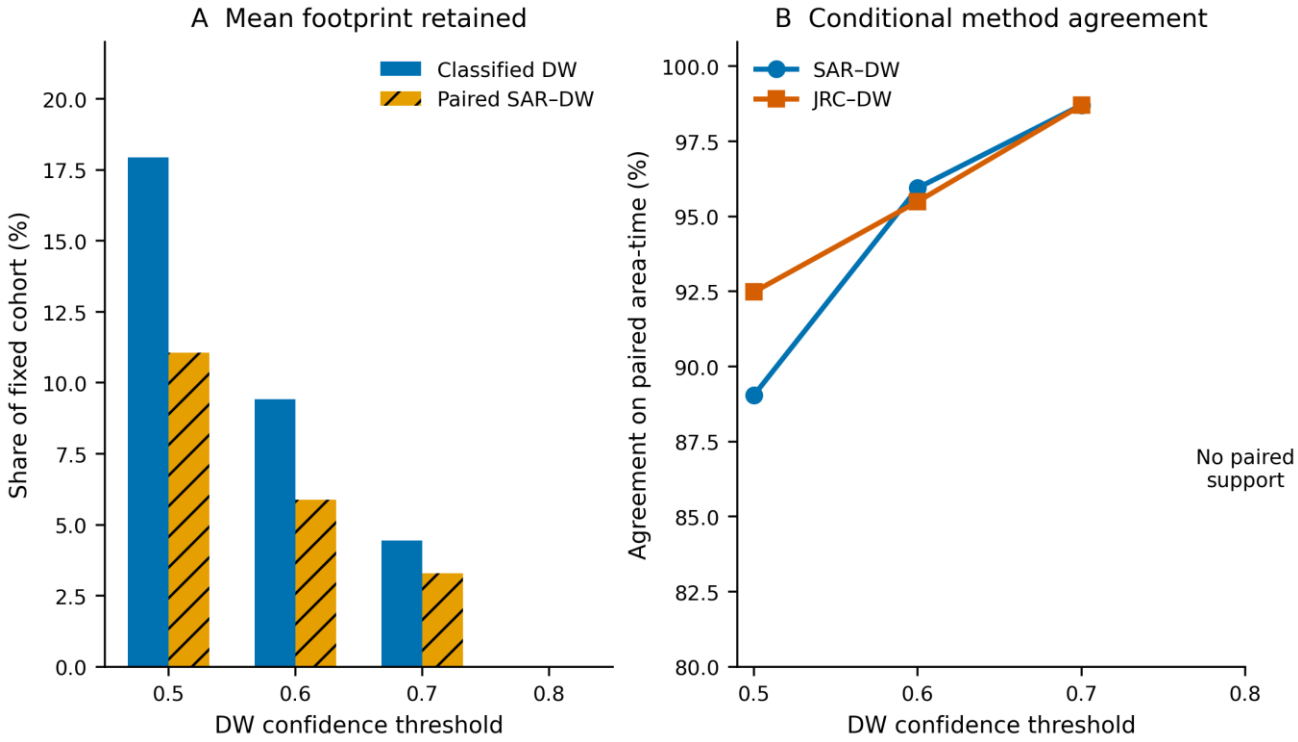


Figure 6. Dynamic World confidence sensitivity on the separate 2015–2024 union historical ever-water cohort. (A) Mean classified and SAR–Dynamic World paired footprint fractions, including months with zero support. (B) Agreement pooled by paired area-time. Threshold 0.8 has no comparison support and no agreement estimate. The changing paired population prevents interpreting the highest agreement as independently validated accuracy. All four thresholds were retained without selecting a best threshold.

JRC–Dynamic World agreement followed the same conditional pattern, increasing from 92.48% at 0.5 to 98.70% at 0.7. The evidence does not identify a best threshold: both the compared population and its size change with the threshold. In particular, 98.71% radar–optical agreement describes the small paired area-time subset, without estimating study-wide accuracy.

The full monthly support and method-envelope series are displayed in Figure 7. The high-threshold ranges remain wide because omitted support enters the upper endpoint; a narrow assigned subset does not constrain the water state of the rest of the cohort.



Figure 7. Dynamic World threshold-specific support and method envelopes, 2015–2024, for the union historical ever-water cohort. Unclassified support enters the upper endpoint; envelopes do not include all physical classification errors.

Radar endpoint regressions used 90 supported monthly observations per region/cohort. Positive time coefficients for both envelope endpoints survived the separate district adjustments for Rajshahi's initial seasonal cohort: 0.0582 logit fraction year^{-1} for the lower endpoint (95% interval 0.0236–0.0929; adjusted p=0.0161) and 0.0396 for the upper endpoint (0.0124–0.0669; adjusted p=0.0312). Other district/cohort combinations were not uniformly supported. These later-period radar cohort outcomes and the primary annual seasonal-class loss concern different estimands; they cannot be interpreted as cancelling trends.

Seasonal support and class-specific diagnostics are shown in Figure 8. At thresholds 0.5 and 0.7, mean paired support during June–September was only 0.313% and 0.110% of the union historical-water cohort, versus 16.45% and 4.87% in other months. A bounded November 2024 score-stage probe examined one land-context area and three stable-source-water interiors. Maximum raw water scores were 0.619 and 0.753, respectively, and monthly-median maxima were 0.506 and 0.748; neither probe retained threshold-0.8 support. These selected areas cannot establish a domain-wide probability ceiling or rule out every processing error. The dated assisted comparison also differed between water and nonwater agreement, particularly in Naogaon. Neither diagnostic is an independent accuracy assessment.

The completed 2018/2024 pilot reproduced all 35 original area columns to within 10^{-6} km² and reconciled every four-cell partition with its paired area. For the union historical ever-water cohort, SAR–Dynamic World pooled agreement increased from 88.92% at threshold 0.5 to 99.45% at 0.7, while paired support

fell from 12.91% to 3.34% of intended area-time. At 0.7, Dynamic World classified 99.29% of the paired area-time as water; symmetric water agreement was 99.72%, compared with 72.14% for nonwater (Figure 8D). For Dynamic World–JRC, threshold-0.7 pooled agreement was 98.06%, water agreement 99.01% and nonwater agreement 42.02%, on 3.89% paired support. Only one of four June–September months in each pilot year had any SAR–Dynamic World pair at thresholds 0.5–0.7. Threshold 0.8 supplied no joint cells. All years, cohorts, seasons, thresholds and directional disagreement cells remain in the archived tables; these selected summary numbers do not extend the joint-cell pilot to the full decade.

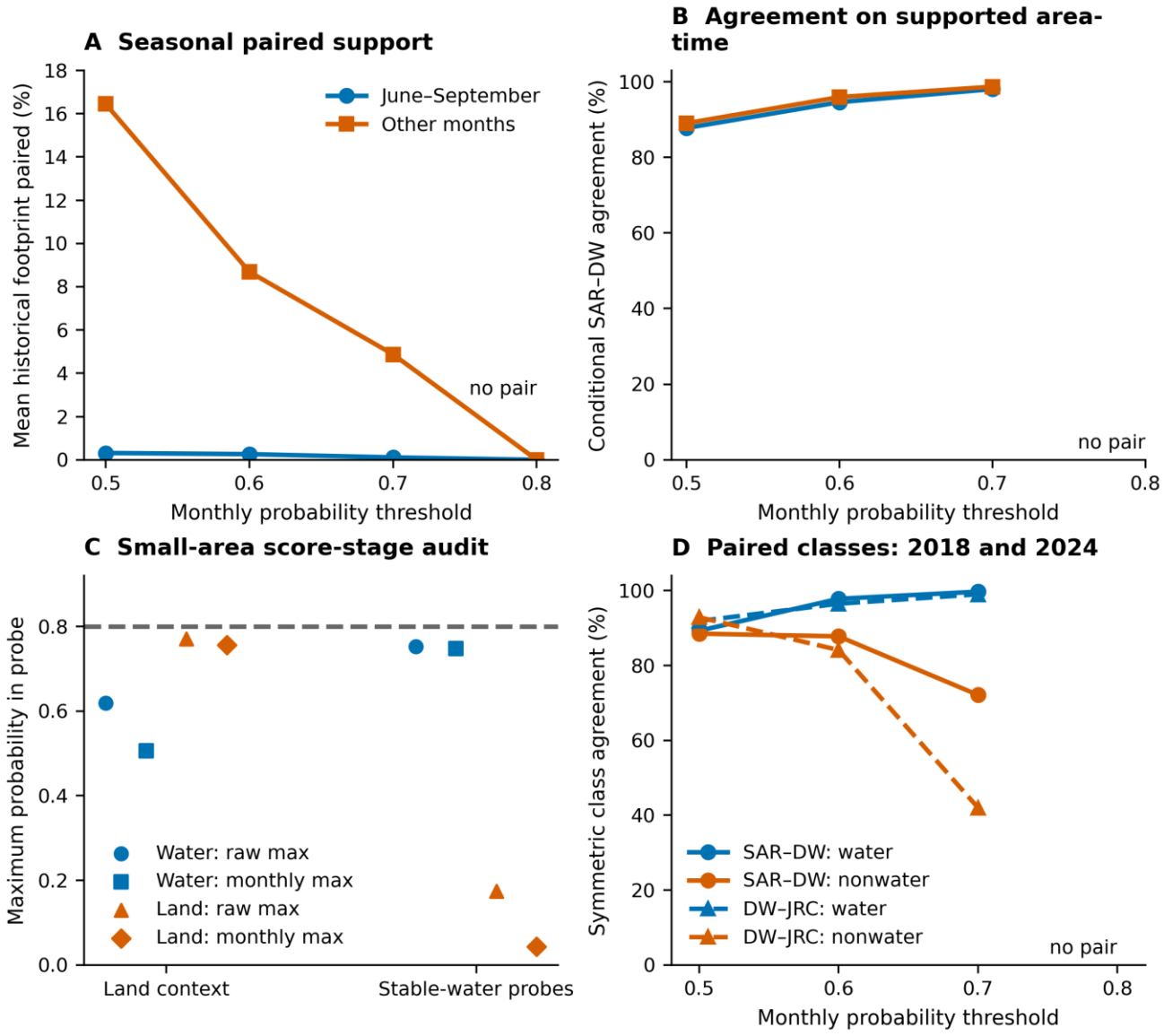


Figure 8. Seasonal and class-specific limits to satellite agreement. (A) Mean SAR–Dynamic World paired support in June–September and other months during 2015–2024, retaining zero-support months. (B) Agreement pooled by paired area-time in those seasons. (C) November 2024 maximum raw and monthly-median probabilities in one small land-context probe and three stable-source-water interiors; probes are selected diagnostics, not independently labeled validation sites. The dashed line is threshold 0.8. (D) Symmetric positive (water) and negative (nonwater) product agreement in the completed 2018/2024 union historical ever-water pilot, pooled across all 24 intended months. Product pairs have different support. Threshold 0.8 has no pairs; its class agreement is unavailable. Panels A–B cover ten years, whereas D covers the two pilot years. No panel estimates independent accuracy.

3.4 Assisted dated evidence leaves a substantial unresolved comparison frame

The 720 sampled pixel-months had 3,022 actual scene-point views from 69 source scenes. Initial triage assigned 337 automatic proxies, selected 245 difficult points for dated human review, and left 138 points

without sufficient dated support. Human review recorded 694 date-specific answers: 430 water, 236 nonwater and 28 uncertain. After preserving date conflicts and the single documented correction, 194 reviewed points had consistent binary dates, 35 showed temporal change and 16 had uncertain dated evidence (Figure 9; Table 3).

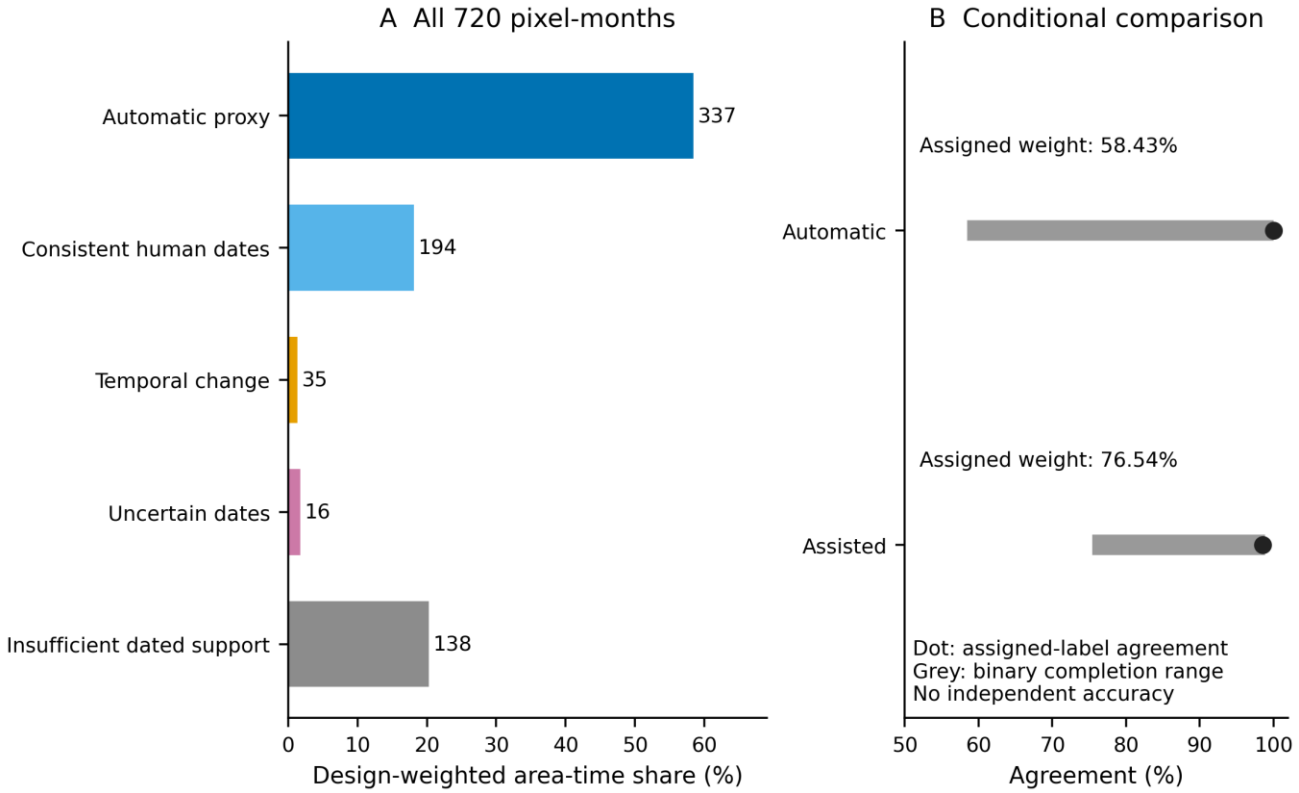


Figure 9. Dated proxy evidence and unknown comparisons for all 720 stratified pixel-month samples. (A) Design-weighted outcome shares; annotations give sample counts. (B) Dots indicate conditional agreement on assigned labels; grey segments show algebraic ranges under arbitrary binary completion of unresolved comparisons. These are not physical-accuracy bounds. Repeated scene dates are within-sample evidence, and selected human review was assisted. Counts and weights use the corrected active label version, with original interpretations preserved.

Table 3. Outcomes and comparison accounting for all 720 sampled pixel-months. In panel A, weight is the share of all sampled area-time for each outcome; in panel B, it is the assigned share for each comparison route. Agreement uses assigned weight only. Completion ranges arbitrarily assign unresolved comparisons to agreement or disagreement; they are neither confidence intervals nor physical-water accuracy bounds. The assisted conditional sampling interval is 96.85–99.87%; the automatic conditional bootstrap is degenerate and omitted. Dashes denote inapplicable quantities.

Outcome or comparison route	Pixel-months (n)	Weight (%)	Agreement (%)	Completion range (%)
A. Outcomes for all sampled pixel-months				
Automatic proxy assigned	337	58.43	—	—
Consistent human dates assigned	194	18.11	—	—
Temporal change: unresolved	35	1.38	—	—
Uncertain dates: unresolved	16	1.80	—	—
Insufficient dated support: unresolved	138	20.28	—	—
Total	720	100.00	—	—
B. Assigned comparison routes				
Automatic only	337	58.43	100.00	58.43–100.00
Assisted proxy	531	76.54	98.56	75.44–98.90

Combining automatic proxies with consistent human dates assigned 531 pixel-months; 189 remained unresolved. Assigned points represented 76.54% of design-weighted area-time, compared with 58.43% under automatic screening alone. Assisted conditional JRC/proxy agreement was 98.56%, with an approximate stratified sampling interval of 96.85–99.87%. The corresponding binary-completion range across all sampled weight was 75.44–98.90% (Table 3). Its width reflects unresolved comparison weight; it is not a confidence interval for true-water accuracy.

The automatic-only assigned subset had no observed JRC disagreement, but 41.57% of design weight remained unassigned. Its conditional bootstrap was degenerate and was not reported as an informative uncertainty interval. Human selection and assistance, shared optical information, repeated dates and the documented feedback correction also preclude an independent-accuracy interpretation. The four selected months do not validate every historical year or the annual distinction between seasonal and permanent classes.

3.5 Drought recovery duration remains unidentified

The meteorological drought history is shown in Figure 10. SPI-3 describes rainfall deficits, so its return to zero does not itself establish surface-water recovery.

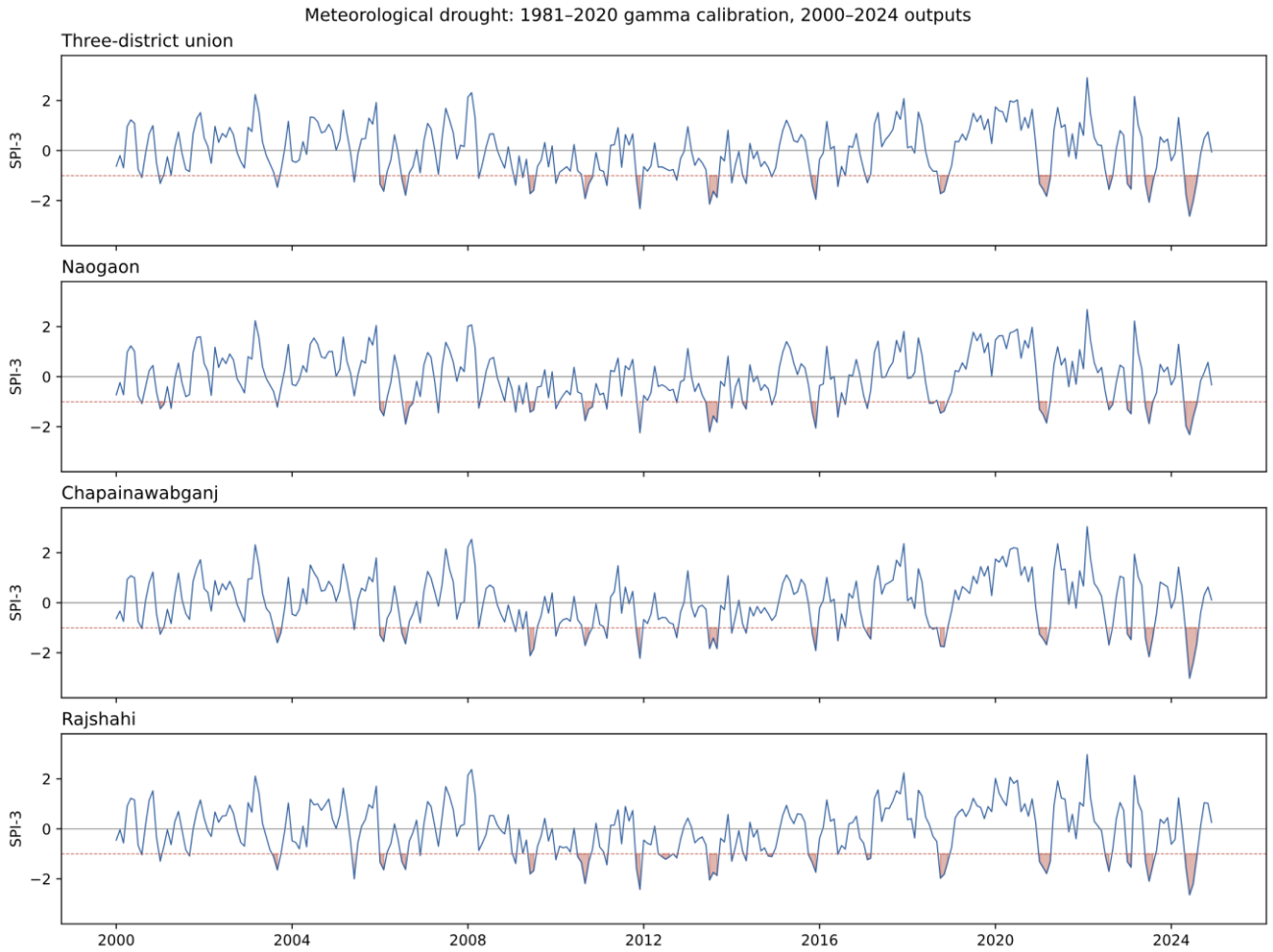


Figure 10. SPI-3 history from CHIRPS version 3 with fixed 1981–2020 calibration. A meteorological drought event is distinct from an observed surface-water deficit.

The monthly JRC observation/reference diagnostic in Figure 11 shows why the original optical record cannot provide a continuous recovery sequence under the stated support rules. The JRC and radar recovery domains remain separate.

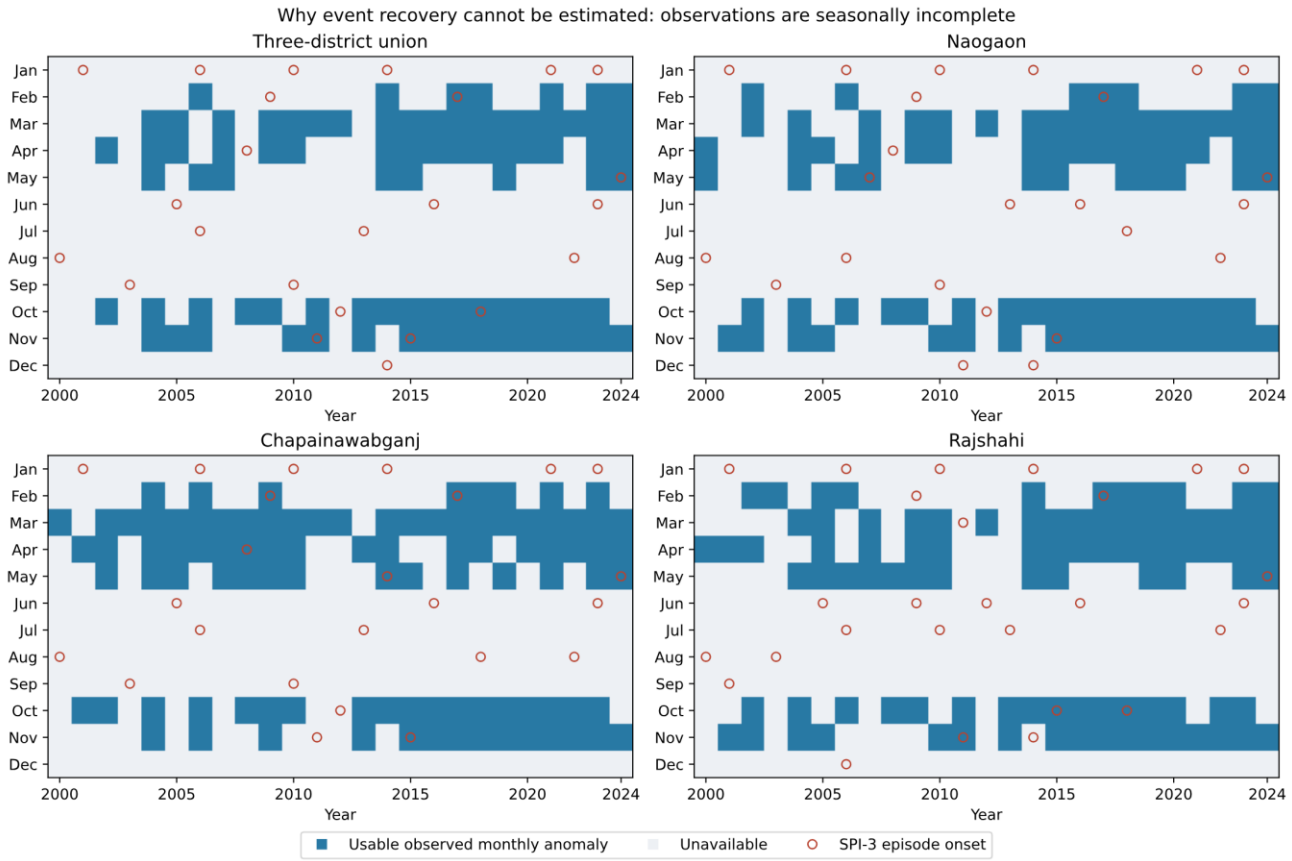


Figure 11. Original monthly JRC water-observation and recovery-gap diagnostics. Observation exclusions remain gaps, and this domain/reference differs from the radar recovery analysis.

Under the primary radar recovery specification, none of the 108 event–cohort records yielded an identifiable duration. Eighty-four failed the pre-event support/reference requirement and 24 failed drought-phase support/reference requirements. Across the nine coverage/reference settings, all 972 records remained without a recovery duration: 702 had pre-event/reference gaps, 144 had drought-phase/reference gaps and 126 had a deficit uncertain within the method envelope (Figure 12).

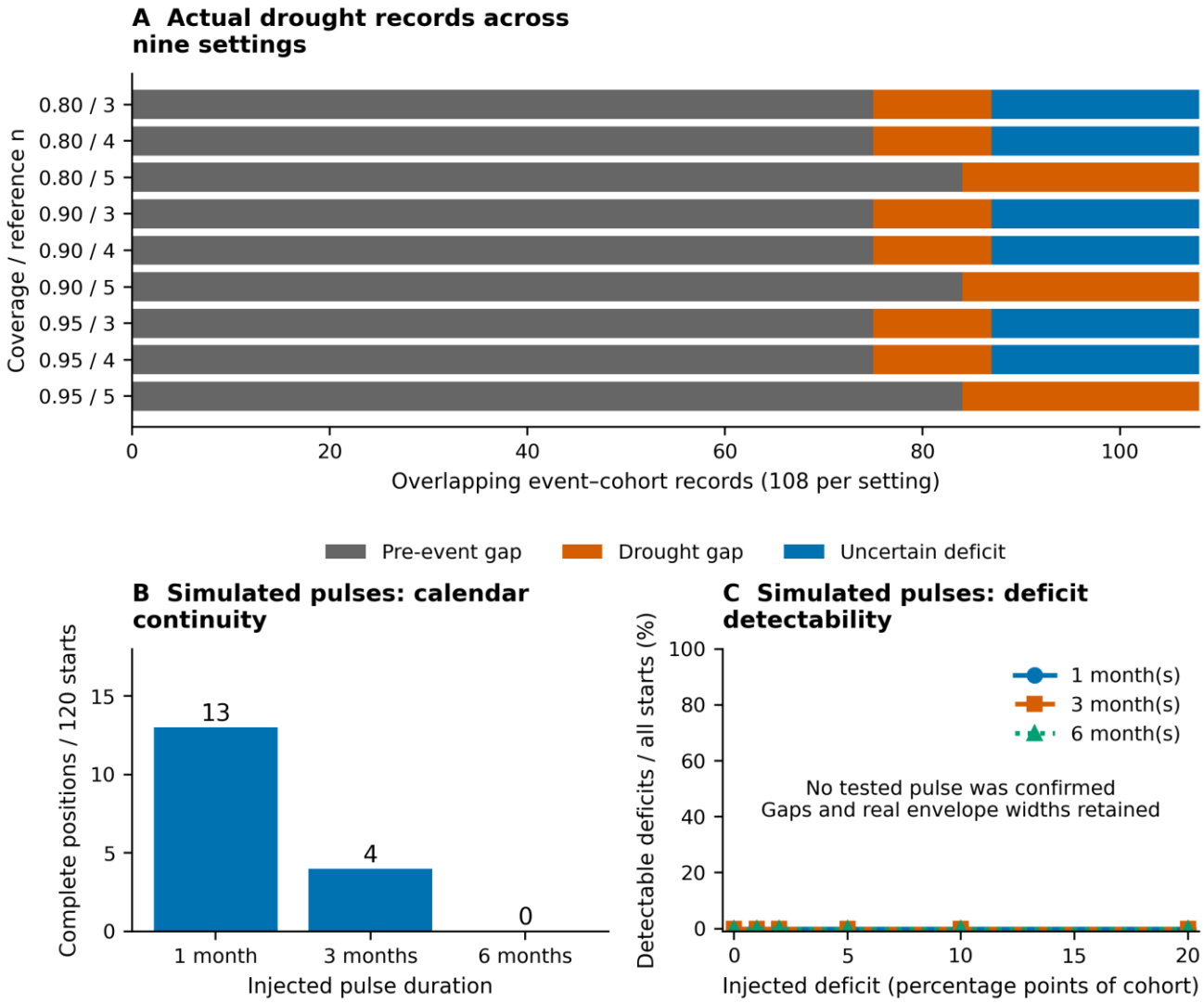


Figure 12. Actual recovery-identifiability failures and separate synthetic detectability diagnostics. (A) Failure status for 108 overlapping event–cohort records at each of nine coverage/reference settings; these are not independent drought counts. (B) Union historical-water pulse positions with complete observations from three months before onset through two confirmation months, out of all 120 possible starts. (C) Confirmed synthetic deficits as a percentage of all starts for three pulse durations; no tested deficit was confirmed. Simulations impose a zero-baseline rectangular deficit with abrupt return, while retaining actual gaps and anomaly-envelope widths. They do not estimate observed recovery duration or validate physical detection power.

No minimum, maximum or scalar recovery time was available under these settings. Relaxing coverage and reference counts changed the reason for non-identification in some records, without producing reliable timing. Because records overlap and observational censoring can be informative, these counts do not estimate the proportion of independent droughts that failed to recover. The finding concerns the inability of the available sequence and method envelope to identify recovery; ecological recovery itself remains unresolved.

Synthetic controls exposed an additional limitation (Figure 12B,C). For the union historical-water cohort, only 13, four and zero of 120 possible starts retained the complete pre-event-to-confirmation sequence for one-, three- and six-month pulses. Across all 25,920 region/cohort/duration/deficit scenarios, no injected deficit was confirmed, including deficits of 0.20 of cohort area. Wide real anomaly envelopes prevented confirmation even where the calendar was continuous. Zero detections for the zero-deficit controls cannot demonstrate specificity when the positive controls also fail. These simulated results explain limited detectability under the declared method; they are not measured drought deficits, recovery times or evidence that real recovery did not occur.

4 Discussion

4.1 Seasonal decline is a monitoring signal with a defined spatial scope

The mapped seasonal decline is supported by more than the choice of two endpoint years, but its magnitude depends on the comparison. The original endpoint decrease was 25.42%, whereas five-year means gave 14.70% on the original footprint and 18.79% on the baseline-water complete cohort. Whole-series associations and these window comparisons support investigating seasonal source-class contraction within retained areas. They do not justify one district-wide loss percentage. District uncertainty also matters: the controlled seasonal models retained clearer evidence in Naogaon and Rajshahi than in Chapainawabganj.

Several mechanisms remain compatible with the observed seasonal-class loss. Water occurrence may have changed, pixels may have moved between seasonal and permanent classes, or the interaction of observation timing and processing may have affected annual assignment. The fixed footprint removes changes in which pixels enter each annual area sum, but it does not remove class error or uneven seasonal sampling. Neither a persistent time coefficient nor a climate-adjusted residual identifies irrigation abstraction, land conversion, river management or an ecological mechanism.

Static landscape context narrows where follow-up could begin. The larger proportional seasonal decrease away from mapped rivers in 2020 cropland was consistent across the tested buffers, while the distant noncropland group gained seasonal-class area. However, a 2020 land-cover label cannot describe cultivation throughout 2000–2021, and a global centerline buffer cannot separate all river water from ponds. River migration, wetland conversion, crop-water management and observation effects remain rival explanations. No local groundwater or discharge time series was acquired to discriminate among them.

Regional evidence of rainfall heterogeneity (Esha et al., 2026) and coupled climatic and groundwater stress (Sarker et al., 2026) makes those mechanisms relevant hypotheses. Testing them would require matched observations of groundwater levels, abstraction, river regulation, land use and waterbody management. Basin-scale separation of storage components uses different evidence, including process models and gravity observations (Prusevich et al., 2025). Our classified areas do not supply equivalent information about water volume. Likewise, the surface-water expansion reported in southwestern Bangladesh (Huda et al., 2021) is compatible with a different regional trajectory; neither result should be generalized across the country without harmonizing domains and definitions.

The permanent class illustrates the risk of interpreting endpoints as a robust process trend. Its 41.04% endpoint increase is a valid description of source classifications on the retained footprint. However, the annual time coefficients lost clear positive support after the 2016 indicator. The appropriate interpretation retains both findings. A continuous increase in physically permanent water, an increase in storage, and improved dry-season water security are not established by this record.

4.2 Agreement should be evaluated together with the area it represents

Raising the Dynamic World threshold removed much of the classification and paired footprint while increasing agreement. This is consistent with selecting clearer, more homogeneous cases, although the present analysis does not isolate the causes of selection. High agreement on that subset cannot describe the omitted mixed, cloudy or weak-probability areas. Threshold 0.8 is particularly instructive: an apparently demanding confidence rule produced no comparison support for this implementation and domain. Treating its missing results as dry land would reverse their meaning.

The completed joint cells identify a further selection effect: the high-threshold paired population was almost entirely classified as water. Thus, strong pooled agreement need not imply comparable nonwater agreement. The two-year pilot separated 99.72% water from 72.14% nonwater SAR–Dynamic World agreement, and 99.01% from 42.02% for Dynamic World–JRC. These are symmetric comparisons between products on changing paired populations, not reference-based class accuracies. Their interpretation depends on the retained area, class prevalence and sparse seasonal support. They do not establish the corresponding class-specific ratios in every year of 2015–2024.

The assisted sample has a related support limitation. The hybrid comparison expanded assigned support beyond the highly selected automatic subset, while some interpretations remained unresolved. Reporting 76.54% assigned area-time alongside 98.56% conditional agreement makes the coverage limitation visible. The binary-completion range additionally shows the influence of unresolved weight without assuming random missingness or inventing labels. It still depends on the assigned proxy labels and cannot account for their possible errors. Accuracy-assessment literature requires reference quality and sample design to be evaluated separately (Olofsson et al., 2014; Stehman and Foody, 2019). Our assisted route meets a different purpose: documenting available evidence and unresolved support.

Cross-sensor studies with external reference imagery (Vanderhoof et al., 2023) and monitoring studies incorporating field observations (Slinski et al., 2025) provide stronger designs for future validation. Their reported performance cannot be transferred to the thresholds or wetland settings used here. The Bangladesh flood-insurance comparison similarly concerns a different outcome, but its finding that data choice affects decision performance (Saunders et al., 2025) supports the practical value of retaining divergence and support information. For district monitoring, agreement should accompany explicit maps or summaries of the population actually compared.

Independent reference evidence remains a separate requirement. A published dataset of expert-delineated, 3 m PlanetScope surface water provides a relevant design example (Mukherjee et al., 2024), but its availability does not establish eligible reference coverage in these districts and months. The attempted provider download returned an access-block page rather than label data; no such files were used. Dates, spatial overlap, class definitions and independence would need verification before applying external labels. The assisted interpretations already collected here were retained honestly rather than relabeled as independent truth.

4.3 Climate association and drought resilience require different evidence

Climate adjustment addressed a limited associational question. The monsoon model improved on its matched mean benchmark but not its trend benchmark; reduced observation/processing models improved on both on only four test years. Those results neither establish robust forecasting skill nor identify a climatic or management cause. Approximate HAC and residual-block intervals, leave-one-year-out fits and simpler specifications expose sensitivity; they cannot compensate for a short series or an imperfect observation process.

Recovery timing requires more than an annual association or several well-interpreted dates. The pre-event baseline, full drought phase and successive recovery observations must be connected on an interpretable calendar. Here, gaps and uncertain deficits prevented that chain under every examined setting. Hydrological drought can persist beyond a rainfall deficit because of storage and seasonal processes (Van Loon, 2015), but this analysis cannot quantify that persistence. It also cannot rank the districts by recovery speed.

The negative identifiability result is useful for monitoring design. Additional observations should be targeted to missing parts of an event sequence, especially pre-event conditions and successive months around climatic drought termination. Reprocessing more isolated clear images may improve an annual map without resolving those event-level gaps. A future recovery study would benefit from a predefined network of sentinel waterbodies with repeated dated imagery and water-level observations, coupled to a stable reference method. Such a network is a proposed next study, not an intervention evaluated here.

4.4 Implications for sustainable water management

For the target districts, the results support a staged use of satellite evidence. Seasonal-class losses on the retained footprint can identify areas for field or higher-resolution investigation. Before using those areas to prioritize storage restoration or water allocation, managers would need to determine whether the signal reflects persistent inundation change, land use, channel movement or classification behavior. The present district aggregates do not locate a validated restoration site or quantify a water-supply benefit.

A practical district dashboard could report four quantities together: source-class area, observation effort, the fraction of the intended footprint with a usable comparison, and the continuity of evidence around drought events. This would preserve the distinction between a complete download, a classified pixel and a continuous hydrological record. The results give empirical reasons for each item: restricted annual coverage, sparse wet-season observations, threshold-dependent paired support and unidentified recovery duration.

This approach is relevant to monitoring the extent of water-related ecosystems and to climate-adaptation planning. It is not a complete official assessment of Sustainable Development Goal indicator 6.6.1, which has its own ecosystem definitions, reporting dimensions and national implementation requirements.[4] Water quality, groundwater storage, biodiversity outcomes, household water access and the effects of management interventions were not measured. The management contribution is therefore better-specified evidence for investigation and monitoring, without claiming observed improvements in water security.

4.5 Limitations and transferability

Retrospective selection is the main spatial limitation. The ever-water footprint uses the full 2000–2024 annual record, and the annual common footprint also requires future-year validity. Transient monthly-only water and poorly observed pixels can be excluded. Equal inclusion of retained pixels across years improves comparability within that subset, but it does not make the subset representative of the entire districts. Raster-center district assignment in the local annual analysis also differs from the radar analysis's geometry-weighted area accounting.

The baseline-water cohorts reduce dependence on future water occurrence, and the 2021-validity alternatives remove the requirement for later annual availability. They do not remove annual-class error, nonrandom missingness or the need for within-year observations. C7 also covers only 1,299.60 km² of annually valid pixels, far below the complete reporting boundaries. The fact that only November passed the equal-calendar retention gate prevents interpreting the monthly check as a fully controlled annual hydroperiod analysis.

Source classification and alignment remain important. Corrected annual products, Landsat collection differences, mixed boundary pixels, SAR threshold behavior, wet soil, rice and vegetation can all affect water estimates. The method envelopes cover specified thresholds and unknown support, while excluding some systematic errors. Dynamic World and the dated optical proxy have related inputs, and monthly sensor composites do not necessarily represent the same day. Independent annual seasonal/permanent accuracy remains unmeasured.

Statistical uncertainty is also incomplete. Twenty-two annual observations provide limited information for models with multiple climate, effort and processing terms. HAC intervals are approximate and pointwise. Reported adjustment families cover specified district tests, not the entire exploratory workflow; spatial dependence and unobserved drivers remain. Retrospective cohorts make the temporal checks conditional hindcasts. The four-month proxy bootstrap addresses sampled pixel-month variation, without resolving nonrandom reference failure or spatial clustering. The lack of independent labels is explicit and prevents presenting proxy agreement as accuracy.

These limitations bound transferability to similar monitoring problems. Estimates should be reported together with their spatial, temporal and reference support. The numerical trends and thresholds themselves remain specific to the declared products, domains and periods in northwestern Bangladesh.

5 Conclusion

Mapped seasonal JRC source-class area decreased within the retained annual-observation footprint in northwestern Bangladesh. The 2000–2021 endpoint decline was 25.42%, compared with 14.70% between five-year means on the same footprint and 18.79% on a baseline-water complete cohort. Negative seasonal associations persisted across the tested specifications, with stronger controlled district evidence

[4] UNEP description and methodology for indicator 6.6.1: https://www.unep.org/en/goal-6-clean-water-and-sanitation/indicator-661.

in Naogaon and Rajshahi. Permanent-class endpoint gains remained sensitive to observation and processing controls. These findings concern classified areas and declared spatial support, not district-wide water storage.

The complementary analyses identify limits that matter for water-management interpretation. Higher radar–optical agreement occurred on a smaller paired footprint; assisted dated interpretation left 23.46% of sampled design weight unresolved; and no drought-recovery duration was identifiable across nine support/reference specifications. Satellite-based climate-resilience assessment should therefore retain observation density, paired support and event continuity alongside mapped change. These results guide further investigation and monitoring design while leaving causal drivers, independent accuracy and ecological recovery unresolved.

Data and code availability

Derived tables, revision protocols, source ledgers and analysis code accompany this revision. The original v0.2.0-review archives retain the earlier statistical replay and are explicitly distinguished from the new revision outputs; they alone do not reproduce the updated models and figures. The revision supplement records the new cohort summaries, model outputs, context and simulated diagnostics, with hashes and execution instructions. Full raw raster histories, large third-party source downloads and dated image crops are not duplicated. Product versions and restrictions are listed in Table 1 and Supplementary Methods. Development materials are stored in the private repository https://github.com/asifahamed11/northwestern-bangladesh-surface-water; access is restricted to authorized accounts. Private repository availability does not by itself satisfy public data-access requirements.

Author contributions

Conflict of interest

The authors report no financial or commercial conflicts of interest. Md. Kamruzzaman and Jayanta Das are Topic Editors of the Research Topic to which this work is intended to be submitted. An independent handling editor is requested.

Funding

The authors report no funding for this research.

Acknowledgments and generative AI use

We acknowledge the organizations providing the water, climate, satellite and administrative-boundary products used in this study. OpenAI Codex, based on GPT-6 (OpenAI; accessed October 2026), assisted with programming, evidence organization, manuscript drafting and editing. Procedural research guidance included Scientific Agent Skills (Kassis et al., 2026), an arXiv preprint and associated software resource. Figures were plotted from the recorded data using conventional scientific plotting tools. Responsibility for scientific interpretation, authorship and the final submitted content rests with the human authors.

Supplementary material

Supplementary Methods retain compact source, grid, label-history and reproduction notes. All twelve figures and three tables are in the main manuscript. The original eighteen main-result and twenty extended diagnostic tables remain available alongside the new revision tables. The completed eight-quarter pilot archive includes unmodified CSV downloads, task IDs, source-snapshot hashes, four joint-cell tables and a portable replay command. The revision manifest identifies current outputs, superseded diagnostics and unresolved independent-reference and recovery questions.

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